

SIH26148

SOFTWARE EDITION

Blockchain & Cybersecurity

Challenge #148 of 236

Creation of scripts/functions with new programming language to commence Computer & Network forensic analysis without triggering security solutions

Sponsoring Organization: National Technical Research Organisation (NTRO)

Theme / Domain: Blockchain & Cybersecurity

Submission Deadline: 30 September 2026

Submitted Ideas: 0/500

Cash Prize: ₹1,00,000 / Winning Team

OFFICIAL PROBLEM STATEMENT BACKGROUND & REQUIREMENTS:

- Background Modern antivirus solutions restrict proprietary software from executing or creating custom scripts designed to analyze the system for deep forensic system analysis. They rely heavily on behavioral heuristics, static signature matching, common compiler outputs (like standard MSVC or GCC artifacts), typical API call sequences and kernel-level monitoring to intercept activities. However, a significant paradigm shift may occur when programmers adopt sophisticated software engineering practices—specifically continuous integration and continuous deployment (CI/CD).
- Description Creating 'Next-Gen' programming language framework, named as 'JOCKY' using cross-platform compiler (windows & ubuntu) which enables systematic creation of scripts for analyzing malicious activities and also provide the complete digital forensics of the computer or network. By utilizing this specific new developed programming language, the framework will not be hindered by any of the existing anti-virus in the environment. This framework should include various scripts/functions which combined with automated polymorphic engines, custom encryption, and multi-vector in-memory execution via native components or Bring your own vulnerable driver (BYOVD) techniques. Framework also able to handle multiple system analysis simultaneously using central management interface. The traffic b/w management interface and client should be routed through trusted cloud infrastructure or content delivery networks (CDNs) using domain fronting or legitimate cloud APIs.
- Expected Solution The scope of the problem is to create scripts/functions in the proprietary programming language (named JOCKY) which enables the user to detect the adversaries:1. Independent programming Language - Programming language or custom Language-independent intermediate representation (LLVM) frontend alters basic control-flow graphs, token generation, and binary structures, rendering signature-based detection ineffective.2. Polymorphism in scripts/function generated - Rather than manually packing a binary, the scripts/function in framework uses a continuous delivery pipeline. Every iteration automatically passes through integrated obfuscators, variable-encryption routines, and polymorphic engines. This ensures that every deployment instance possesses unique hashes, modified entry points, and altered import tables, neutralizing traditional file-reputation databases.3. Living-off-the-Land & BYOVD Execution - The scripts/functions in framework should avoid standard, noisy API calls for core operations like persistence, privilege escalation, and network routing (SOCKS5). Instead, it relies on:A. In-Memory Execution: Utilizing multiple distinct file-less techniques (e.g., process hollowing, reflective DLL injection, API unhooking, direct system calls, or thread execution hijacking) to run secondary script entirely within the memory space of trusted processes.B. Kernel-Level Subversion: Detection of legitimate or vulnerable third-party drivers (BYOVD) to disable EDR callbacks or manipulate kernel structures directly, blinding security agents running in user or kernel space.